Toronto KSI Collision Hotspot Dashboard

Project 2 Report — MBAN 5140 | Data Visualization and Storytelling

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1. Overview

This report documents the “Top 20 Collision Hotspots in Toronto (2006–2023)” Tableau dashboard — an interactive analytical tool built from data sourced through the City of Toronto’s Open Data Portal. The central decision question: Where and when do the most dangerous Killed or Seriously Injured (KSI) collisions occur, and which contributing factors are associated with them?

The intended audience is Toronto city executives and public safety planners who require actionable, location-specific intelligence to allocate traffic enforcement resources, redesign high-risk intersections, and evaluate the long-term impact of road safety policy.

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2. Data Source

The dataset — Toronto KSI Collision Records (2006–2023) — contains 18,957 records spanning 6,872 unique collision events across Toronto where at least one person was killed or seriously injured. Key fields include: collision date and time, intersection name, geographic coordinates, road classification, injury type, involved party (pedestrian, cyclist, motor vehicle), and four binary contributing-factor flags — Aggressive Driving, Alcohol Involved, Red Light Infraction, and Speeding. The full dataset covers all Toronto intersections; this dashboard narrows focus to the Top 20 hotspot intersections, which account for 134 KSI collisions in the current default view. Contributing-factor flags are not mutually exclusive; a single collision may be flagged for more than one factor, and each BAN tile reflects the flagged-factor share within the current filter selection.

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3. Dashboard Design

The dashboard contains six purpose-built components, each answering a distinct sub-question for the audience, arranged in a top-left to bottom-right reading path.

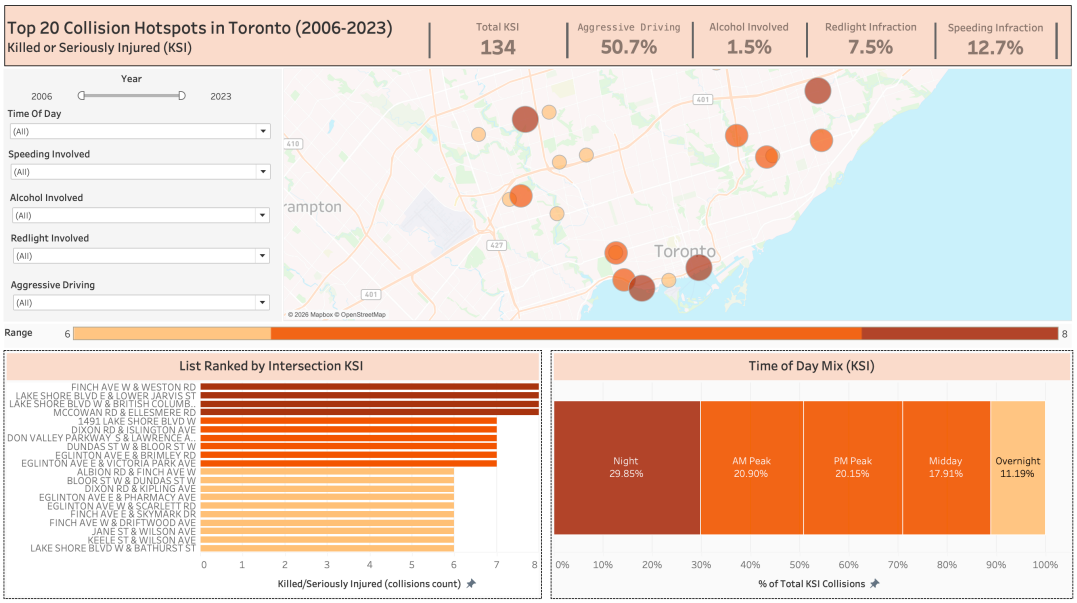

Figure 1 — Default view: Top 20 KSI Collision Hotspots, Toronto (2006–2023).

3.1 Header + BAN KPI Tiles

Four BAN tiles at the top of the dashboard surface the most executive-relevant outputs: shares of KSI collisions involving Aggressive Driving (50.7%), Alcohol (1.5%), Red Light Infractions (7.5%), and Speeding (12.7%), alongside total KSI count. All tiles update dynamically with every filter change and map or list selection.

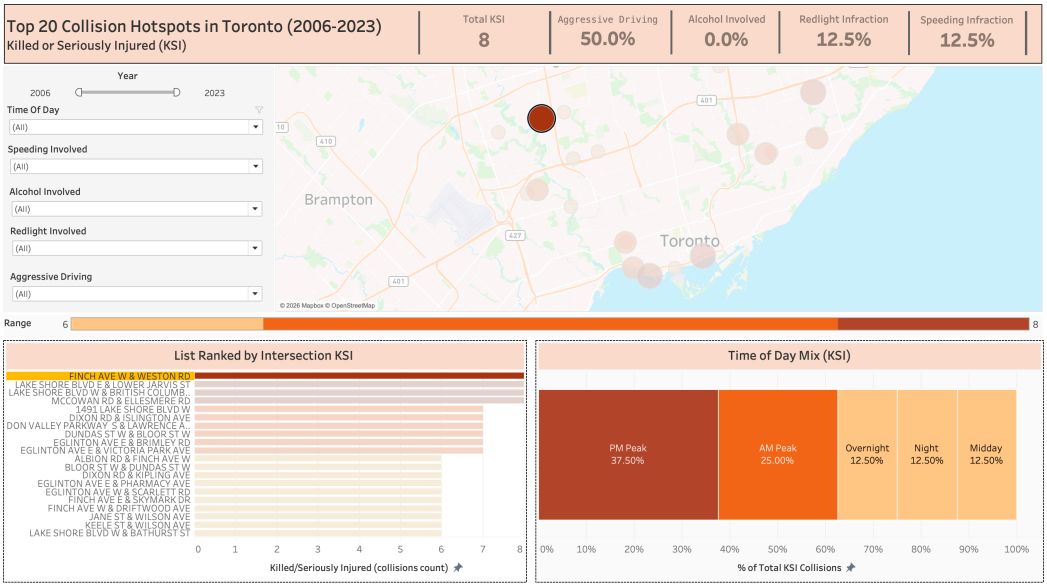

Figure 2 — Hotspot drill-down: selecting Finch Ave W & Weston Rd filters all BANs and the Time of Day Mix to intersection-level data; values shown differ from the dataset-wide defaults quoted above.

3.2 Interactive Filter Panel

Six filters — Year range slider (2006–2023), Time of Day, and four contributing-factor toggles — drive simultaneous cross-filtering across all views, enabling users to isolate time periods or specific risk factors and observe how the entire dashboard responds.

3.3 Geographic Hotspot Map

A proportional symbol map plots the 20 most dangerous intersections; circle size and color intensity encode KSI collision counts per intersection, allowing planners to identify spatial clustering along major arterials. Selecting any hotspot filters all other views to that intersection's data.

3.4 Top 20 KSI Ranked List

A horizontal bar chart ranks the 20 most dangerous intersections by cumulative KSI count (2006–2023), sorted descending for legibility; bar length encodes the same measure as the map, and the view is bidirectionally action-connected to it.

3.5 Time of Day Mix (Stacked Bar Chart)

A 100% stacked bar chart decomposes KSI collisions across five time-of-day periods — Night (29.85%), AM Peak (20.90%), PM Peak (20.15%), Midday (17.91%), and Overnight (11.19%) — providing planners a temporal profile for scheduling enforcement patrols.

4. Key Findings

Finding 1 — Aggressive Driving Is the Dominant Contributing Factor (50.7%)

Aggressive driving is flagged in 50.7% of KSI collisions at the Top 20 hotspot intersections — more than all other contributing factors combined. This is not a marginal result; it indicates that behavioral factors are associated with the majority of Toronto's most severe collisions at these locations. The policy implication is direct: targeted enforcement of aggressive driving (dangerous lane changes, tailgating, street racing) at these specific intersections represents the primary intervention opportunity for city planners.

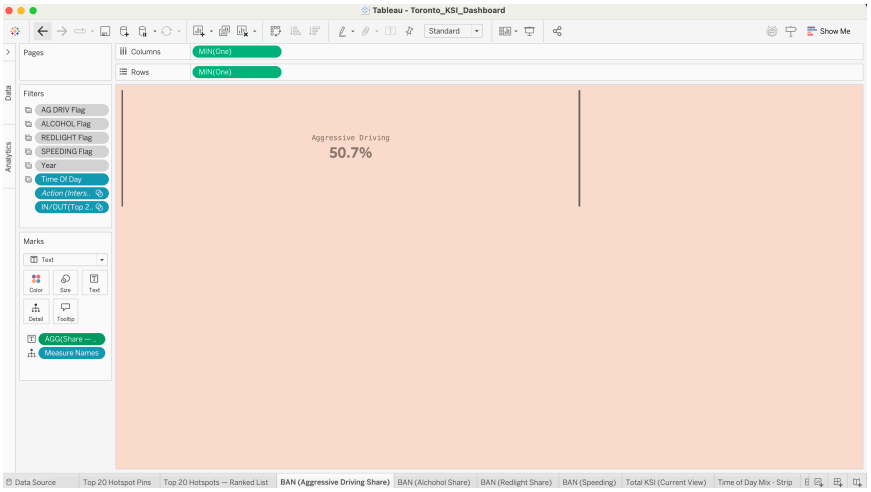

Figure 3 — Aggressive Driving Share BAN at Top 20 hotspot intersections.

Finding 2 — Night Hours Carry the Largest Risk Share (29.85%)

The Night period (6 PM–midnight) accounts for 29.85% of KSI collisions — the largest share of any time window. AM Peak (20.90%) and PM Peak (20.15%) confirm elevated commuter-hour risk. Even the Overnight window (midnight–6 AM), despite lower traffic volumes, accounts for 11.19% of KSI events — suggesting that fatigued or impaired late-night driving remains a persistent risk at the city's most dangerous intersections, independent of volume.

Finding 3 — Secondary Contributing Factors Are Present but Less Prevalent

Alcohol is flagged in only 1.5% of Top 20 hotspot KSI collisions, consistent with the sustained effectiveness of Toronto's RIDE programs; sober but aggressive and speeding drivers represent a far larger risk at the city's highest-risk intersections. Red-light infractions are flagged in ~7.5% of events (about 1 in 13), supporting the case for expanding Automated Speed Enforcement and red-light camera coverage at the ranked hotspot intersections — a cost-efficient intervention with documented deterrence effects.

Finding 4 — City-Wide Scale: 6,872 KSI Collisions Over 17 Years

Across all of Toronto, the dataset records 6,872 unique KSI events between 2006 and 2023 — approximately 404 per year. The Top 20 hotspot intersections represent a disproportionate

concentration of risk relative to Toronto's hundreds of signalized intersections, confirming that targeted intervention at a small number of locations can meaningfully move the city-wide KSI metric. Finch Ave W & Weston Rd consistently ranks as the single most dangerous hotspot across multiple filter configurations.

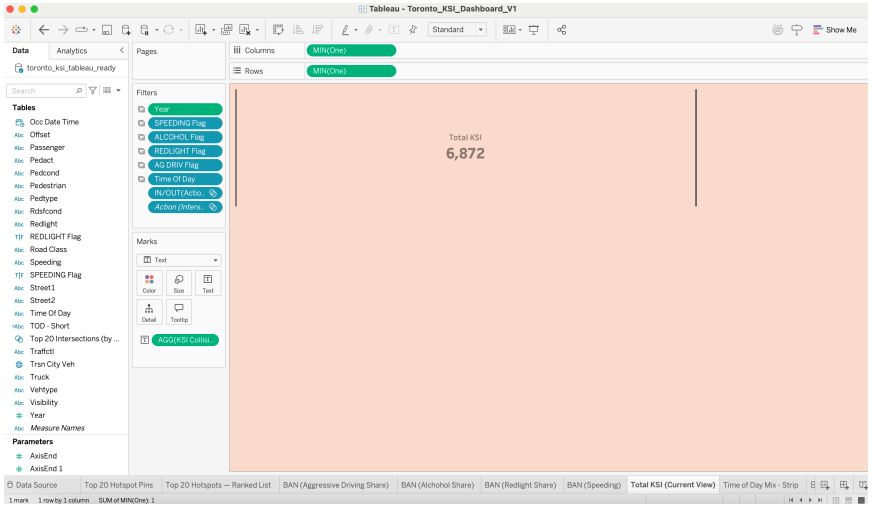

Figure 4 — Total KSI BAN: 6,872 events city-wide (2006–2023).

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5. Design Best Practices Applied

The layout follows a top-left to bottom-right reading path, placing executive-relevant BANs first and supporting analysis below. Color is constrained to a single orange-to-brown gradient encoding severity. Six cross-filters drive simultaneous updates across all views. Tooltips surface intersection-level context on hover, and all unnecessary gridlines and decorative elements were removed to maximize signal clarity.

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6. Conclusion

The Toronto KSI Dashboard transforms 18,957 raw collision records into an executive-ready analytical tool. Aggressive driving is the primary flagged factor in Toronto's most severe collisions (50.7%), nighttime hours carry disproportionate risk (29.85%), and a short list of intersections accounts for a concentration of city-wide KSI events. These findings translate directly into enforcement schedules, red-light camera deployments, and intersection redesign budgets. The most dangerous places in Toronto are known, the risk factors are measurable, and this dashboard ensures decision-makers have the intelligence required to act.

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7. References

City of Toronto. (n.d.). *Motor vehicle collisions involving killed or seriously injured persons* [Data set]. City of Toronto Open Data Portal. https://open.toronto.ca/dataset/motor-vehicle-collisions-involving-killed-or-seriously-injured-persons/